On Boundaries of Evidence

Boundary-State Calculus for Typed Observation, Admissible Transfer, and Falsifiable Persistence

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The problem

Scientific claims often fail not within either of two descriptions, but in the passage between them: interior state to boundary data, one instrument to another, a fine model to a coarse model, a learned operator to a dynamical claim, or a topological invariant to a physical quantity. The paper develops Boundary-State Calculus (BSC) as a typed audit calculus for this passage.

BSC is not proposed as an ontology, a unified physical theory, or a proof of holography, quantum gravity, consciousness, or QCD. Its unit of evaluation is one claimed transfer. That transfer must state what moves, which observables remain meaningful, what information is destroyed, whether target equations remain satisfied, which diagrams fail to commute, what physically implements the passage, and what evidence licenses the result.

System and boundary discipline

The paper transcribes the BSC corpus's original finite-system tuple and its later measurable repair rather than replacing them retrospectively (§3). The hash-identified internal sources are not independently inspectable in the public release, so this lineage is source-local and cannot be replayed by the reader. The repaired system separates raw state from exact observational equivalence and includes controls, outputs, physical context, legal filtration, dynamics, observation, comparison geometry, viability, transformation registries, and a boundary ledger. A companion record supplies reconstruction data, provenance, and certificate state without silently changing the inherited tuple. Instrument, controller, observer or reference frame, clock, and calibration are physical configuration variables, not metadata.

An ideal trace, finite sensor layer, boundary response, reconstruction relation, and selected extension are kept distinct (§5). A boundary response supports an interior claim only relative to a governing equation, coefficient class, equivalence or gauge class, regularity, data regime, and stability theorem.

The transfer record

The central added object is (§7)

$$\mathfrak{M}_{\ell\to m} = \left(T_{\ell m}, T^{\sharp}_{\ell m}, K_{\ell m}, R_{\ell m}, \Theta_{\ell m}, \delta_{\ell m}, C_{\ell m}, \mathsf{Cert}_{\ell m}\right).$$

$T_{\ell m}$ transports states; $T^{\sharp}_{\ell m}$ pulls target observables in the reverse direction. $K_{\ell m}$ is post-processing or a simulation channel. $R_{\ell m}$ records target-equation failure. $\Theta_{\ell m}$ is a declared naturality defect. $\delta_{\ell m}$ is directed Blackwell–Le Cam deficiency: the best error when the source experiment simulates the target. Zero deficiency means arbitrarily accurate simulation; exact Blackwell simulation additionally requires an attaining kernel or an applicable randomization theorem. $C_{\ell m}$ records carrier, controller, instrument, reference frame, clock, resources, and boundary conditions. $\mathsf{Cert}_{\ell m}$ binds assumptions, tolerances, sources, proof or execution artifacts, hashes, open obligations, and status.

For a declared claim, admissibility requires a well-formed and evaluated record, within-tolerance upper enclosures, true hard gates, and adequate readiness. A straddled tolerance is inconclusive, not a proved violation. Residual, observation, deficiency, quotient-loss, and viability coordinates remain typed separately (§8); a missing carrier or pullback is blocked, not a small error.

Simulation evidence. The v1.2.0 profile separates statistical, numerical, and surrogate evidence while binding intended use, typed losses and gates, joint-law semantics, factored identity, and compatibility reserves. If $\ell^1_j \leq \ell^0_j + \rho_j$, admission requires $U^0_j + \rho_j \leq \tau_j$, without recounting uncertainty in U^0_j. Its fixed-interface corollary uses trace distance, total variation, and a POVM boundary: processing cannot resurrect lost distinctions, and normalized postselection needs a separate bound. Energy ports localize residuals, while yields and efficiencies retain denominators and failures. Bernoulli information, one-identity alignment, and four July 2026 probes remain source-bound; neither 137 nor full quantum closure follows.

Electromagnetic evidence. The module `framework/Electromagnetic_Evidence_Bridge.md` (BSC-EM-01–BSC-EM-11) keeps gauge, Maxwell, constitutive, port, inverse, coupling, and metrological claims separate. If $\|\widehat{S} - S\|_2 \leq \varepsilon$ for a power-normalized report, $\sigma_{\max}(\widehat{S}) - \varepsilon > 1$ falsifies that model, $\sigma_{\max}(\widehat{S}) + \varepsilon \leq 1$ certifies contractivity on the declared band, and the middle case is inconclusive. The passive pair $S_0(\omega) = r$, $S_\tau(\omega) = re^{-i\omega\tau}$ has equal power but different delay. Field rescaling and flux quantization do not select the coupling; revised SI gives $\mu_0 = \alpha\frac{2h}{ce^2}$, while RG flow still needs a scheme, scale, thresholds, particle content, and boundary value. For patterned media, translation-faithful materialization gives $\mathrm{Stab}_{\mathrm{tr}}(\kappa) \subseteq \mathrm{Stab}_{\mathrm{tr}}(P)$; otherwise the selector may erase the tiling. Periodic Hat point diffraction and the 2026 centroid-SiN chiral experiment keep aperiodicity sample- and report-local. No BSC theorem derives or predicts $\alpha^{-1} = 137.035\,999\,177(21)$.

Composition and persistence

For compatible partial stochastic transfers, state kernels compose only on supported intermediate laws; observable pullbacks compose in reverse. Observation defects have a contraction-weighted bound, deficiencies a triangle inequality, and equation residuals require a compatible equation map (§7.1; §15). Incompatible interfaces make composition undefined. Persistence is a prefix property of induced path laws (§9), with identity, viability, residual, deficiency, and certificate budgets. Recursive boundary-state maintenance is a target-relative principle, not an established universal physical law.

Provenance and status

- **Imported mathematics.** Markov kernels and operational quotients; Blackwell–Le Cam comparison; boundary trace and Calderón-type response; Koopman residual methods and stated learnability limits; periodic homogenization; sheaf descent; structured cospans; non-invertible defects; and proof checking. These are not claimed as new BSC discoveries.
- **Paper-level consequences.** Assumption-bound results include quotient and exact-decision descent; paired state–observable probability preservation; defect, deficiency, and deterministic residual propagation; simulation-evidence typing, compatibility-bounded deployment, and coupled-surrogate and fixed-interface operational-channel propagation; no-resurrection, energy-port and efficiency, Bernoulli-information, and same-identity corollaries; normalized-scale rate decomposition and covariance; singular-set and slice visibility; analytic zero transfer; the zeta-specific tail, pointwise exponent/rate split, ambient and fixed-β singular sets, contour count, and local root drift; prefix persistence; conditional structured-cospan composition; readiness caps; and the topology-to-charge block. The full quantum-morphism composition theorem remains open. No empirical status is inherited.
- **Open work.** Open problems include universal persistence, empirical validation, stochastic residuals, closure under composition for the full quantum variant (BSC-QOP-03), reconstruction with instrument uncertainty, QRF descent, certified recurrence, a coherent decoration functor for BSC, physical charge bridges, a complete analysis of quantum advantage for zeta algorithms, temperature and origin bridges, and a machine-checked kernel. Their statuses remain conjectural, untested, unexecuted, blocked, or open.

Ten fixtures

The exact fixtures (§16) are:

1. torus winding $(2, -3)$, with electric charge blocked without a physical bridge;
2. distinct one-dimensional conductivities with one Dirichlet-to-Neumann map;
3. binary experiments with directed deficiencies 0 and $1/4$;
4. Koopman residual $\sqrt{2/N}$ separated from polluted zero eigenvalues;
5. two invariant, correlated Bell states with common reduced state $I/2$ whose finite $\mathbb{Z}_2$ frame transforms reduce to orthogonal outputs;
6. locally nonempty parity constraints with no global section;
7. the off-shell residual m^2c of a massive-field shift;
8. an exact receipt for $x = -1$ refuting $\sqrt{x^2} = x$;
9. a finite zeta–coherence identity while leaving the broader framework and application theorems unexecuted and blocking exact-zero, thermodynamic-DQPT, RH, and physical-origin promotions; and
10. one $1/100$ surrogate error stays within tolerance in a stable $a = 1/2$ host but first violates it at step 7 in a stable $a = 9/10$ host.

Fixtures 1–7 and 9 have mathematical derivations but no separate execution receipts. Fixtures 8 and 10 have deterministic CPython receipts. F10 is one-coordinate finite-recurrence code verification, not full admissibility or physical validation. This remains a preprint with audit artifacts, not a complete implementation.

Normalized scales and zeta–DQPT

A certified scale family keeps finite systems, interscale comparisons, ideal carriers A_N, nonzero normalizers Z_N, normalized observables $L_N = A_N/Z_N$, and estimator laws distinct. The framework proves normalization collapse, additive logarithmic rates, normalization covariance, lower-margin rate stability, exceptional-set and slice visibility, contour/root transfer, and total-variation decision bounds. It does not manufacture an infinite-system state or physical phase.

For $s = \beta_{\text{eff}} + it$, the engineered finite model gives the exact identity

$$Z_N(\beta_{\text{eff}})\mathcal{L}_N(\beta_{\text{eff}}, t) = -S_N(s),$$
$$S_N(s) = \sum_{n=1}^{N} (-1)^{n+1} n^{-s}.$$

Abel summation yields

$$|\eta(s) - S_N(s)| \le N^{-\sigma}\left(1 + \frac{|s|}{\sigma}\right),$$
$$\sigma = \Re s > 0.$$

Together with a certified nonzero boundary margin, this gives a bounded Rouché zero-count transfer. It does not turn fitted finite minima into exact zeros, a finite recurrence into a thermodynamic singularity, selected zeros into RH, β_{eff} into the 305 K laboratory temperature, or an engineered encoding into an independent physical origin. Indeed, for fixed $0 < \beta_{\text{eff}} \le 1$, the raw normalized $\mathcal{L}_N \to 0$ for every t. For $0 < \beta_{\text{eff}} < 1$, its fixed-$s$ decay exponent is $1 - \beta_{\text{eff}}$ off the eta zero set and 1 on it. With $N = 2^d$, this proves the source's pointwise rate values $(1 - \beta_{\text{eff}}) \log 2$ and $\log 2$, respectively. The rate discontinuities are exactly the zeta zeros, with jump $\beta_{\text{eff}} \log 2$; confinement of all such points to $\beta_{\text{eff}} = 1/2$ is equivalent to, not a proof of, RH. At a fixed multiplicity-m zero, the m finite partial-sum roots localize at scale $N^{-\beta_{\text{eff}}/m}$, with an explicit signed leading displacement for a simple zero. A fixed-β real-time slice sees exactly the zero ordinates on that line. No uniform-in-height or finite-size empirical conclusion follows. The renormalized limit is $Z_N\mathcal{L}_N$.